China's Missile Silo Construction, 2019-2021

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Topical Report No.3 under DOS 19AQMM20P2156 (08/03/2021-08/10/2021)

10 August 2021

APPROVED FOR PUBLIC RELEASE; DISTRIBUTION UNLIMITED

REPORT DOCUMENTATION PAGE			Form Approved OMB No. 0704-0188

1. REPORT DATE	**2. REPORT TYPE**	**3. DATES COVERED** *(From - To)*
08/10/2021	Geospatial Analysis Report	2019-2021

4. TITLE AND SUBTITLE	
China's Missile Silo Construction, 2019-2021	**5a. CONTRACT NUMBER** 19AQMM20P2156
	5b. GRANT NUMBER
	5c. PROGRAM ELEMENT NUMBER 1033020075, $3,000
6. AUTHOR(S) Babiarz, Renny (author, image analyst) AllSource Analyst Network (imagery analysis, peer review)	**5d. PROJECT NUMBER**
	5e. TASK NUMBER
	5f. WORK UNIT NUMBER
7. PERFORMING ORGANIZATION NAME(S) AND ADDRESS(ES) AllSource Analysis, 1067 Hover St, Unit E, #110, Longmont, CO 80501	**8. PERFORMING ORGANIZATION REPORT NUMBER** IR 20210810-CH
9. SPONSORING/MONITORING AGENCY NAME(S) AND ADDRESS(ES) Department of State/Arms Control, Verification, and Compliance Bureau Office of Assistant Secretary, HST Room 5888 2201C Street, NW Washington, DC 20520	**10. SPONSOR/MONITOR'S ACRONYM(S)** DOS/AVC
	11. SPONSOR/MONITOR'S REPORT NUMBER(S) AVC-VTN-20N03

12. DISTRIBUTION/AVAILABILITY STATEMENT
Public Release Authorized; Distribution Unlimited

13. SUPPLEMENTARY NOTES

Archived at DTIC 04/22/2025.

14. ABSTRACT
Analysis of multiple forms of geospatial data of both missile silo and wind electricity generation infrastructure corroborates open-source reporting of missile silo construction near Jilantai, Yumen, and Hami, China. This suggests that China is expanding deployment of land-based ballistic missile launch systems, which could improve the survivability of China's land-based nuclear missile force. Additional imagery analysis of construction methods, as well as ongoing monitoring of these new missile silo areas, is expected to yield additional insights into their organization and role in meeting China's strategic operational objectives.

15. SUBJECT TERMS
China, PRC, nuclear weapons, Jilantai, Hami, Yumen, missile silo, wind turbine, construction

16. SECURITY CLASSIFICATION OF:			**17. LIMITATION OF ABSTRACT**	**18. NUMBER OF PAGES**	**19a. NAME OF RESPONSIBLE PERSON**
a. REPORT	**b. ABSTRACT**	**c. THIS PAGE**			Rongsong Jih, COR
UNC	UNC	UNC	SAR	14	**19b. TELEPHONE NUMBER** *(Include area code)* 202-647-8126

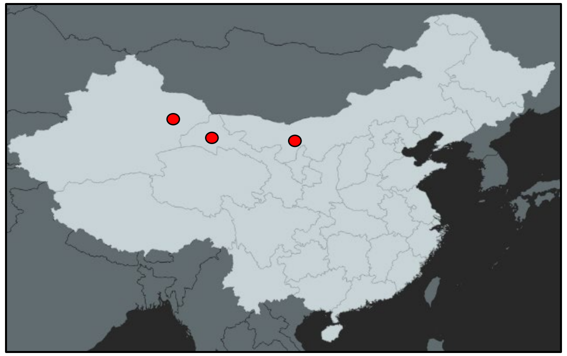
Table of Contents

List of Figures

Location:
China

Coordinates:
35.0000 105.0000

Date of Report:
10 August 2021



Overview

Analysis of multiple forms of geospatial data of both missile silo and wind electricity generation infrastructure corroborates open-source reporting of missile silo construction near Jilantai, Yumen, and Hami, China. This suggests that China is expanding deployment of land-based ballistic missile launch systems, which could improve the survivability of China's land-based nuclear missile force. Additional imagery analysis of construction methods, as well as ongoing monitoring of these new missile silo areas, is expected to yield additional insights into their organization and role in meeting China's strategic operational objectives.

Background

The Department of State's Bureau of Arms Control, Verification, and Compliance (AVC) requested that AllSource Analysis conduct an independent satellite imagery analysis of reported missile silo construction areas in China to resolve disputes related to their function. Multiple media organizations reported that China constructed probable missile silos at three areas between 2019 and 2021: Jilantai, Yumen, and Hami (Warrick; Broad and Sanger; **Figure 1**). A Chinese government-associated online publication reportedly suggested these areas are not missile silos, but rather areas set aside for wind electricity generation (Lendon and Gan). To resolve disputes about the function of these areas, AllSource compared satellite imagery analysis of these areas with known construction indicators of wind electricity generation areas, hereafter referred to as wind turbines.

AllSource Analysis uses the term "air dome" to describe air-supported structures observed during the course of this study. Air domes are composed of fabric supported by air pressure, usually with some limited rigid framing in some areas of the structure.

Analysis

AllSource Analysis compared and contrasted the construction indicators of wind turbines with the more specialized construction indicators observed at reported missile silo construction areas. The following compared the infrastructure and construction processes observed at reported missile silo construction in three areas (Jilantai, Yumen, and Jilantai; **Figure 1**) with those observed at the Yumen wind electricity generation area.

Construction Processes at Jilantai Missile Base are Similar to Yumen and Hami Probable Missile Fields

Analysis of geospatial information showed that construction processes of compact missile silos first identified at a known People's Liberation Army Rocket Force (PLARF) missile training base at Jilantai (Kristensen) are similar to construction patterns observed at both Yumen and Hami probable missile fields. China variously positioned both inflatable and non-inflatable structures over construction areas at all three locations, which serve both to conceal aspects of missile silo-related construction, and provide a degree of climate control that could support construction of environmentally sensitive infrastructure. Additionally, AllSource observed that construction materials, processes, supporting infrastructure, and distance between missile silo construction areas at all three locations were distinct from those associated with wind turbine construction areas.

At Jilantai Missile Base -- a known missile training area for the PLARF – China constructed at least 17 missile silos, mostly between 2019 and 2021 (**Figure 2**). During construction, AllSource consistently observed uninflated air dome fabric covering construction areas; probable metal framing to support air dome fabric; covered, circular probable excavated areas; and inflated air domes covering construction areas (**Figure 3**). In at least one instance, a metal building with a peaked roof was constructed over a missile silo construction site instead of an air dome. Additionally, according to analysis of a Maxar image accessed via Open Street Map, China may have conducted missile silo loading operations at that same missile silo – to include an open silo door, an elevated crane, and a transporter with rails adjacent to the silo -- although AllSource Analysis cannot independently confirm the date of this image, nor that the observed operation involved either a missile or a dummy training model (**Figure 4**). On average, the distance between recently constructed missile silos at Jilantai measure approximately three kilometers.

At the reported Hami missile field, China constructed at least 30 probable missile silos between during 2021; at this time, commercially available satellite imagery precludes a more detailed accounting of probable missile silos in this area (**Figure 5**). During construction, AllSource consistently observed possible air dome-related construction materials; probable metal framing to support air dome fabric; covered, circular possible excavated areas; and inflated air domes covering construction areas (**Figure 6**). Additionally, at one construction area AllSource observed two possible temporary rails on either side of the covered, circular possible excavated area; these could support a gantry crane for moving objects to and from the possible excavated area (**Figure 6**). On average, distances between the probable missile silos at Hami measure approximately 2.7 kilometers.

At the reported Yumen missile field, China constructed at least 110 probable missile silos during 2021 (**Figure 7**). During construction, AllSource consistently observed inflated air domes covering most construction areas, and non-inflatable buildings covering some construction areas (**Figure 8**). Although available commercial satellite imagery precludes more detailed analysis of construction methods, there is currently no indication of any wind turbine electricity generation construction processes or infrastructure in the reported Yumen missile field area. On average, the probable missile silos are separated by a distance of approximately 2.7 kilometers.

Construction Process at the Yumen Wind Electricity Generation Area

The Yumen wind electricity generation area is located in China’s Gansu Province approximately 16 kilometers southwest of Yumen city, 40 kilometers west of the Yumen Plutonium Reprocessing Facility, and 25 kilometers east of the recently reported Yumen probable missile silo field. Given this location, the Yumen Wind Electricity Generation Area is a logical candidate for comparison with construction patterns at the reported Yumen probable missile silo field.

Imagery analysis of construction processes at the Yumen Wind Electricity Generation Area shows distinct construction materials, patterns, and resulting infrastructure that are not hidden from remote observation during construction. Analysis of Maxar satellite imagery during 2010 of wind turbine construction at the Yumen Wind Electricity Generation Area reveals a multistep process: excavate and pour a concrete pad foundation, install rebar framing and a cylindrical turbine base, pour concrete over the rebar framing, gather the turbine support column and fan blades, assemble the wind turbine, install the electricity collection system (e.g. transformer), and connect electricity transmission lines (**Figure 9**). (Note: the construction process described here is an overview of a more complex process.) Further, electricity generation and transmission equipment such as transformers and electricity transmission lines are key requisite signatures of infrastructure supporting wind electricity generation areas; in particular, transformers have not been observed at missile silo construction areas. Additionally, the area’s wind turbines are separated by a distance of 250 meters, versus the 2.7-3 kilometer separation between aforementioned probable missile silo construction areas.

Analysis of multiple wind electricity generation areas in China supports this general construction process characterization, which differs from aforementioned missile silo construction processes. Additionally, AllSource Analysis has not observed any instance of wind turbine construction in China to be concealed from observation beneath other structures as has been observed at the probable missile silo construction sites.

Conclusion

AllSource Analysis found no indication of wind electricity generation construction processes or related supporting infrastructure at the PLARF Jilantai missile base or the reported Yumen and Hami missile silo fields. This analysis did find that construction techniques of compact missile silos and supporting infrastructure at Jilantai are consistent with construction observed at Hami and Yumen, corroborating open-source reporting that China is expanding its land-based ballistic missile launch capability. Given its possible association with China’s deployed nuclear weapons launch capability, these developments should be compared with those of other nuclear weapons-related facilities in China to understand how these fit within China’s nuclear weapons research, development, and production programs, and to better understand their implications with respect to China’s strategic operational objectives.

ALLSOURCE ANALYSIS

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China | 10 August 2021

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Discovery Report

Figure 1
Missile Base Overview, China

35.0000 105.0000

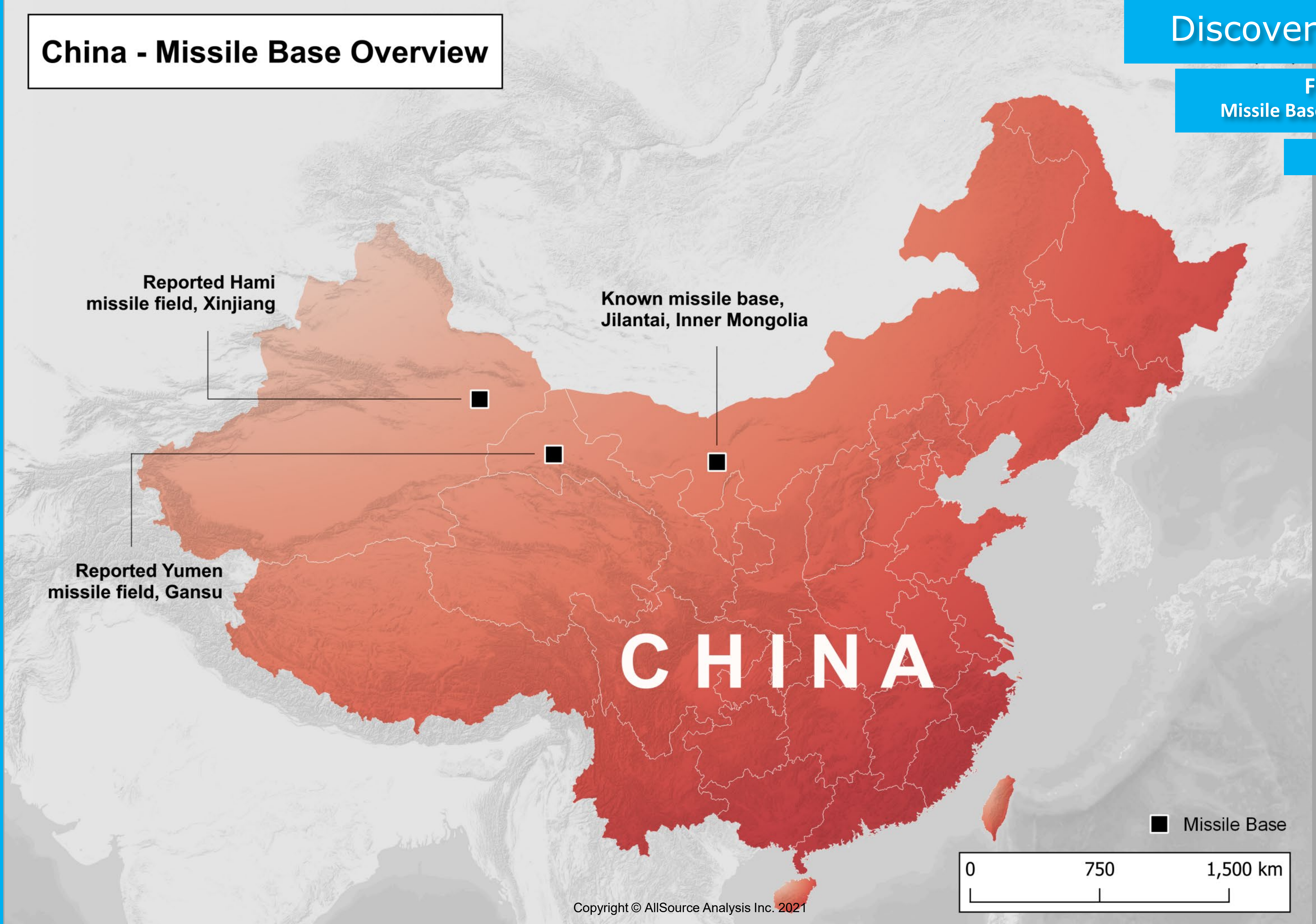

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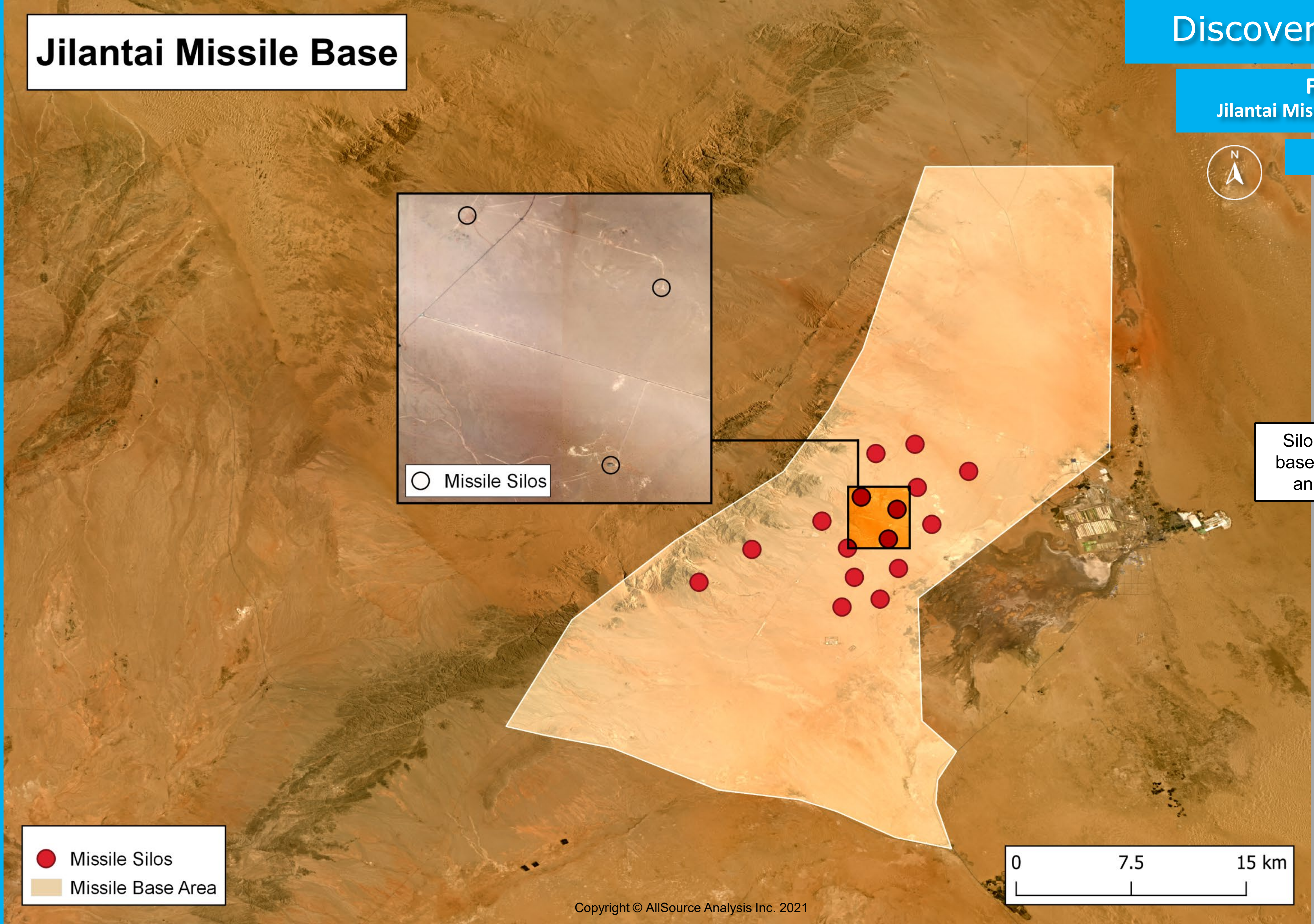
Figure 2
Jilantai Missile Base Overview

39.7471 105.5430



Silo locations estimated based on media reporting and imagery analysis

China's Missile Silo Construction, 2019-2021

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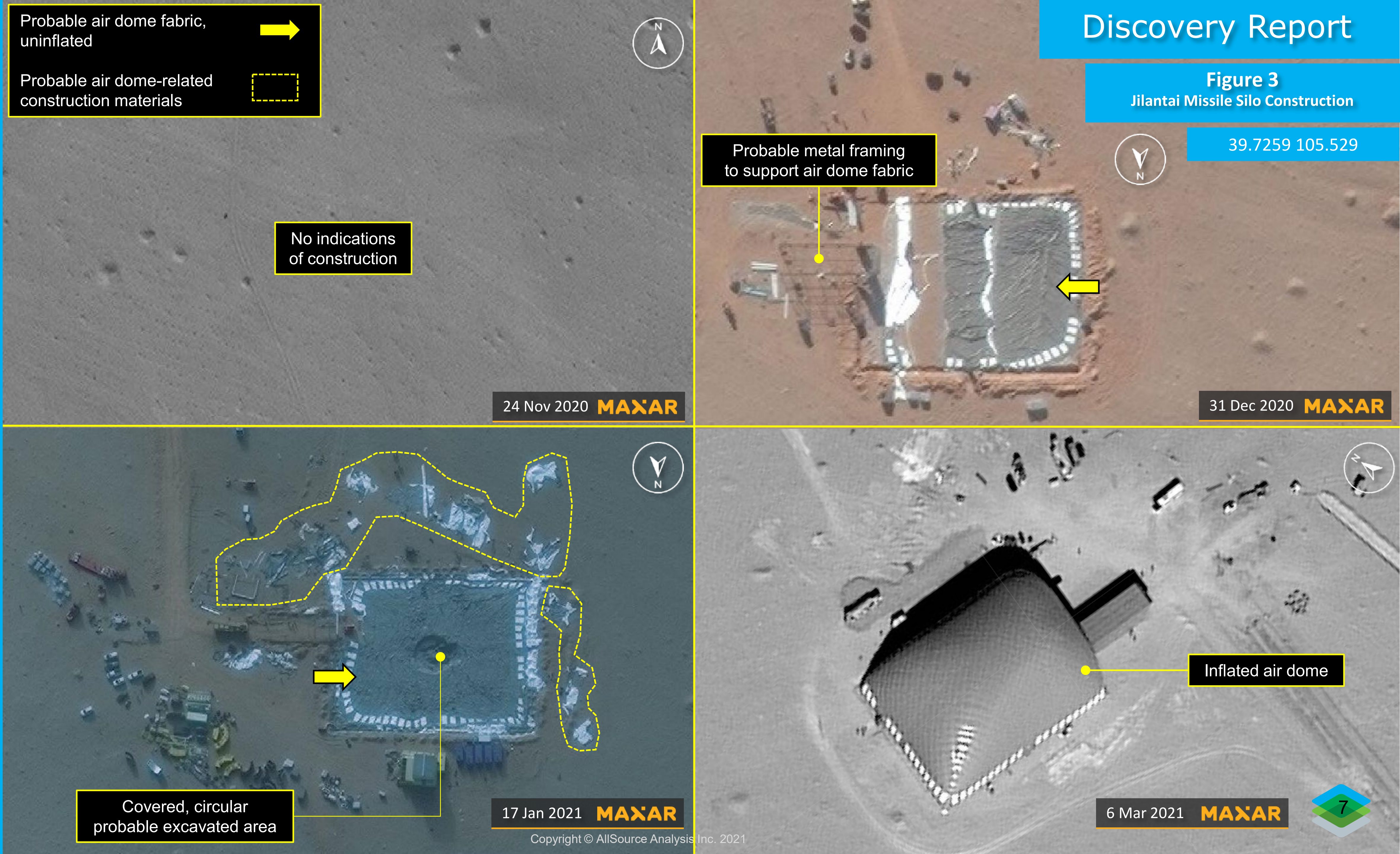
Figure 3
Jilantai Missile Silo Construction

39.7259 105.529



China's Missile Silo Construction, 2019-2021

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Figure 4

Jilantai Missile Silo Operation

39.7646 105.5393

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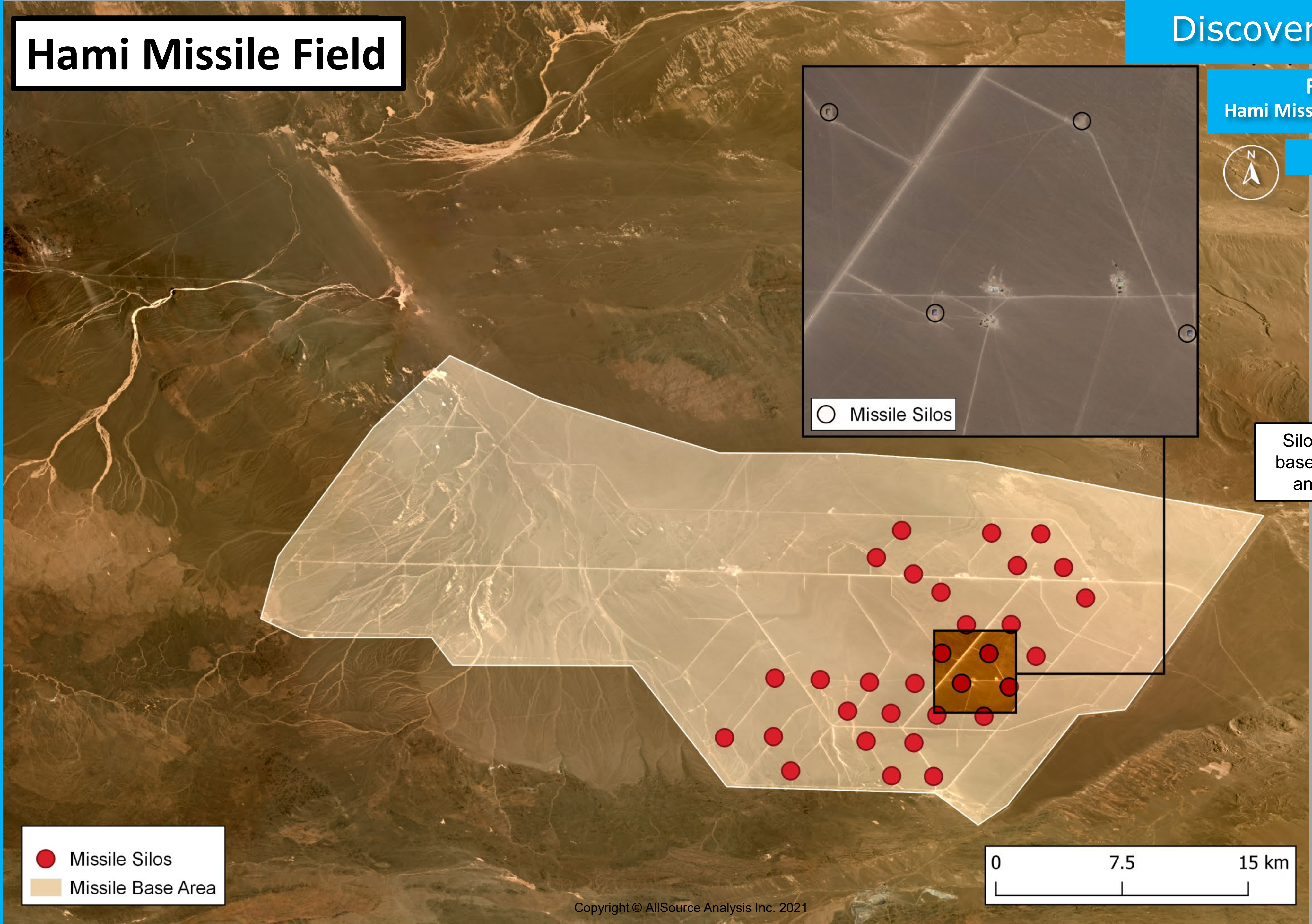
Figure 5
Hami Missile Field Overview

42.3393 92.5174



Silo locations estimated based on media reporting and imagery analysis

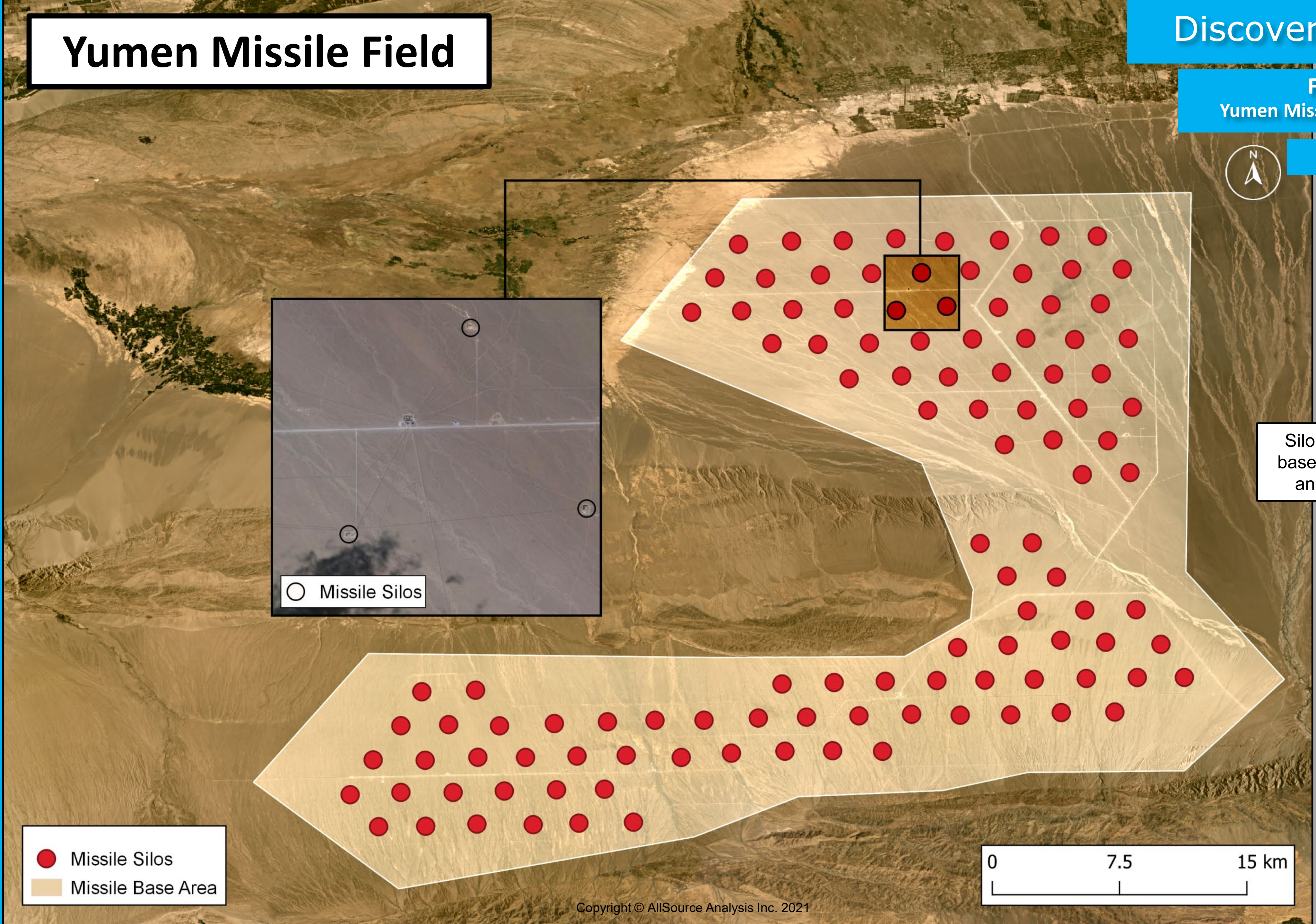
Figure 6
Hami Probable Missile Silo Construction

42.2685 92.6989

Yumen Missile Field

Figure 7
Yumen Missile Base Overview

40.13435 96.5985



Silo locations estimated based on media reporting and imagery analysis

Figure 8
Yumen Probable Missile Silo Construction

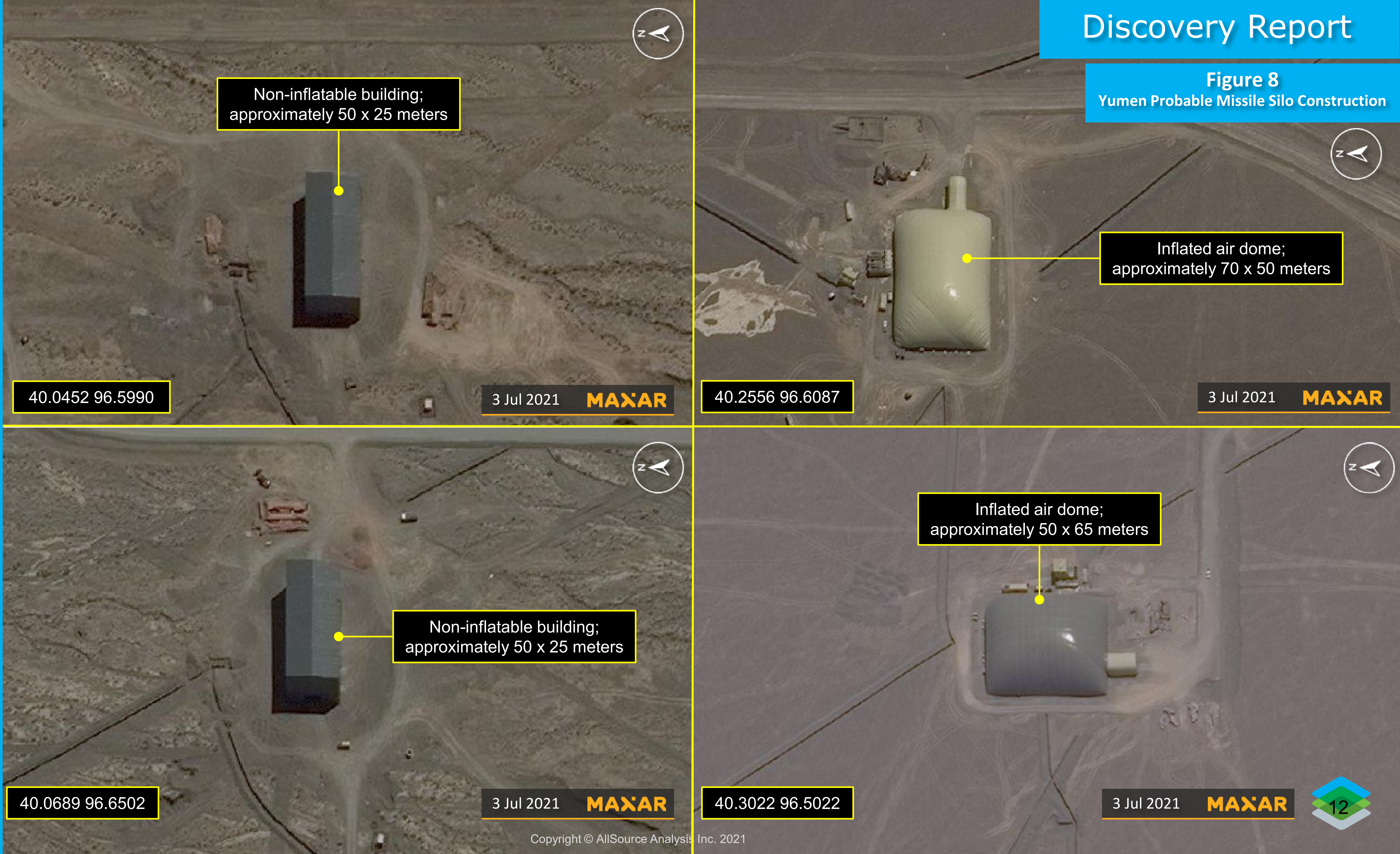
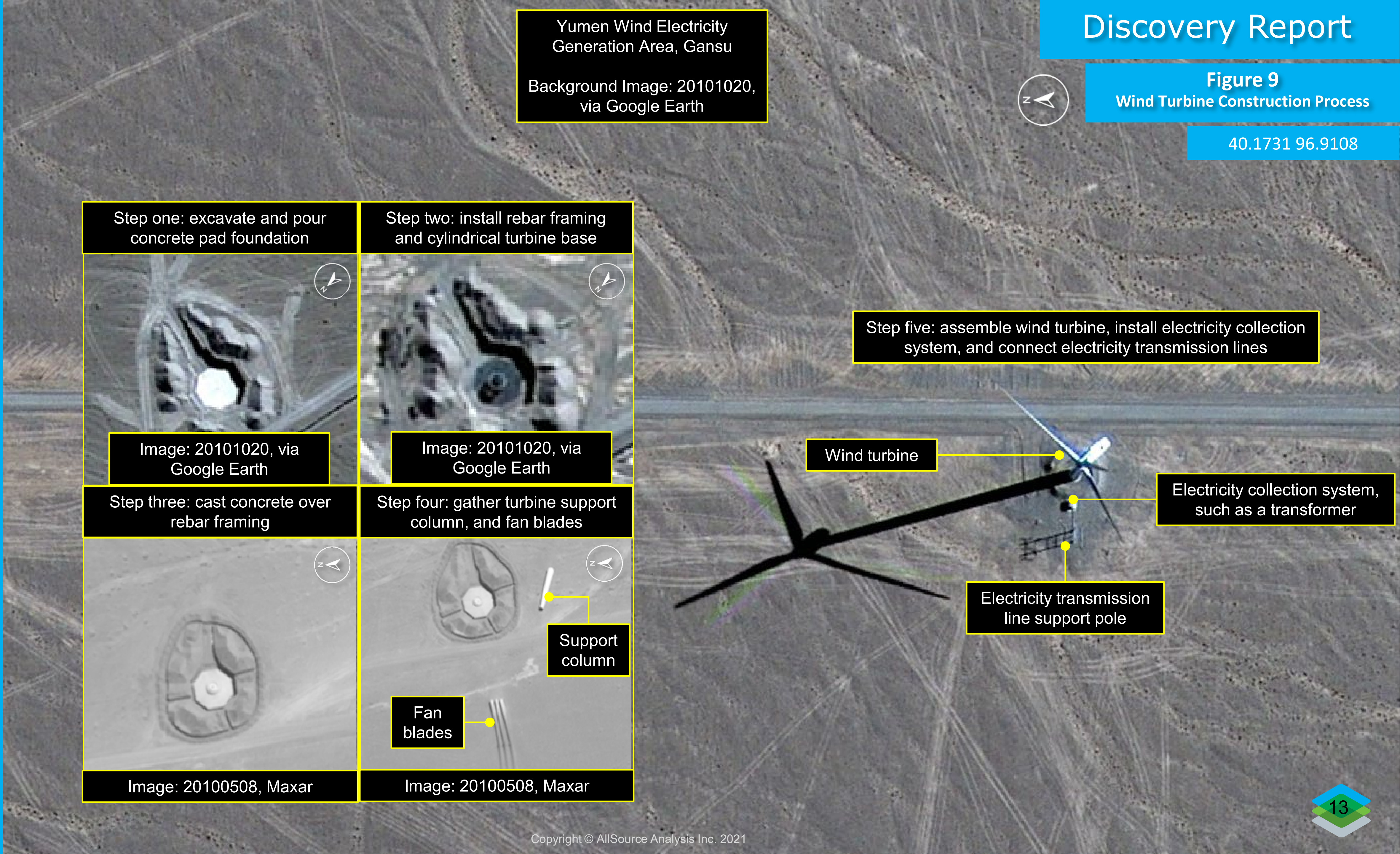


Figure 9
Wind Turbine Construction Process

40.1731 96.9108



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Additional Notes: Estimative Language

AllSource analysts use estimative language to express uncertainties associated with analytic observations and analytic judgements in their products. Analytic observations are any object identifications based on analysis of geospatial information, usually based on satellite imagery analysis. Analytic judgements refer to interpretations of observations or sets of observations as broader statements. The specific estimative language used corresponds to the United States government's Office of the Director of National Intelligence (ODNI) Intelligence Community Directive (ICD) 203.

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